

Lista 3 - Redes de Computadores I

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Exercícios Resolvidos

1) Identificando seu IP:

Abra o terminal da sua máquina e identifique seu número de IP. No Windows, use o comando ipconfig e, no Linux, ifconfig.

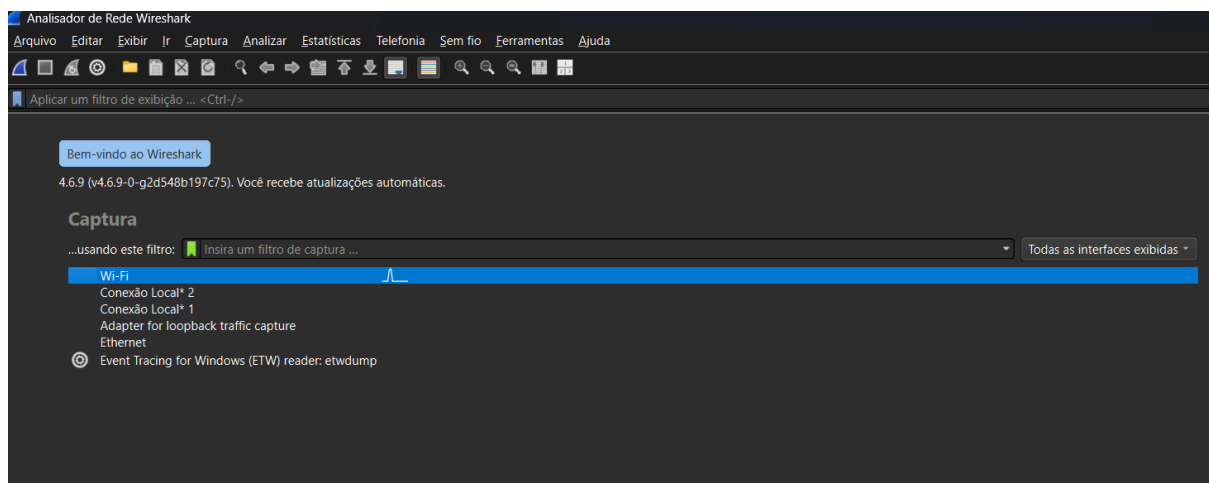
R:

```
Adaptador de Rede sem Fio Wi-Fi:

Sufixo DNS específico de conexão. . . . . : 
Endereço IPv6 de link local . . . . . : fe80::8438:c84c:d998:46ce%11
Endereço IPv4. . . . . : 192.168.255.10
Máscara de Sub-rede . . . . . : 255.255.255.0
Gateway Padrão. . . . . : 192.168.255.1
```

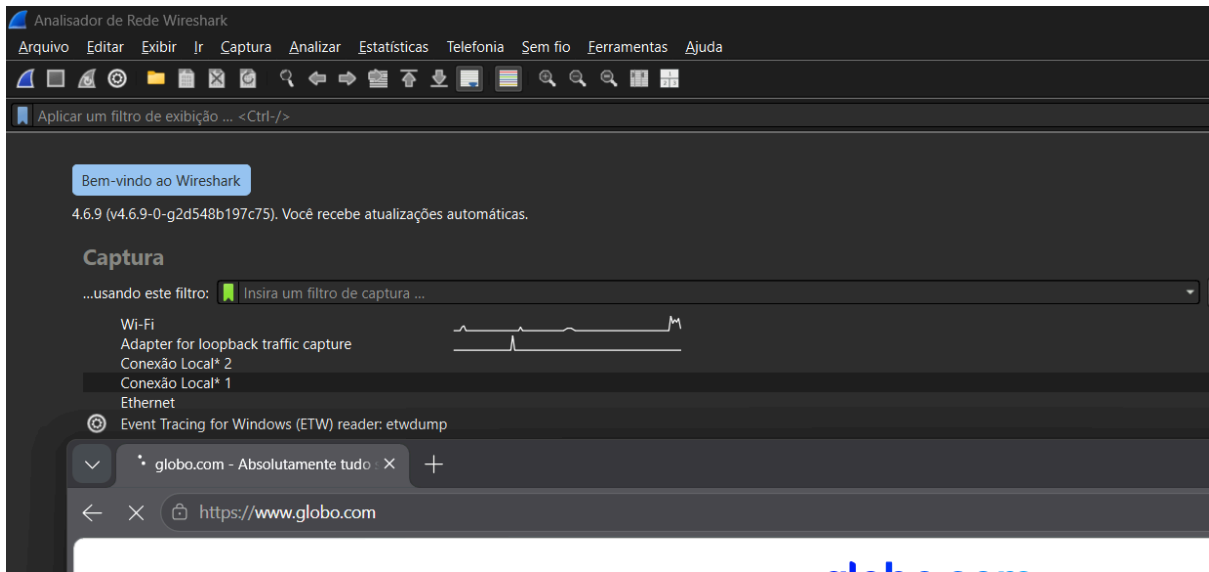
2) Tela Inicial:

R:



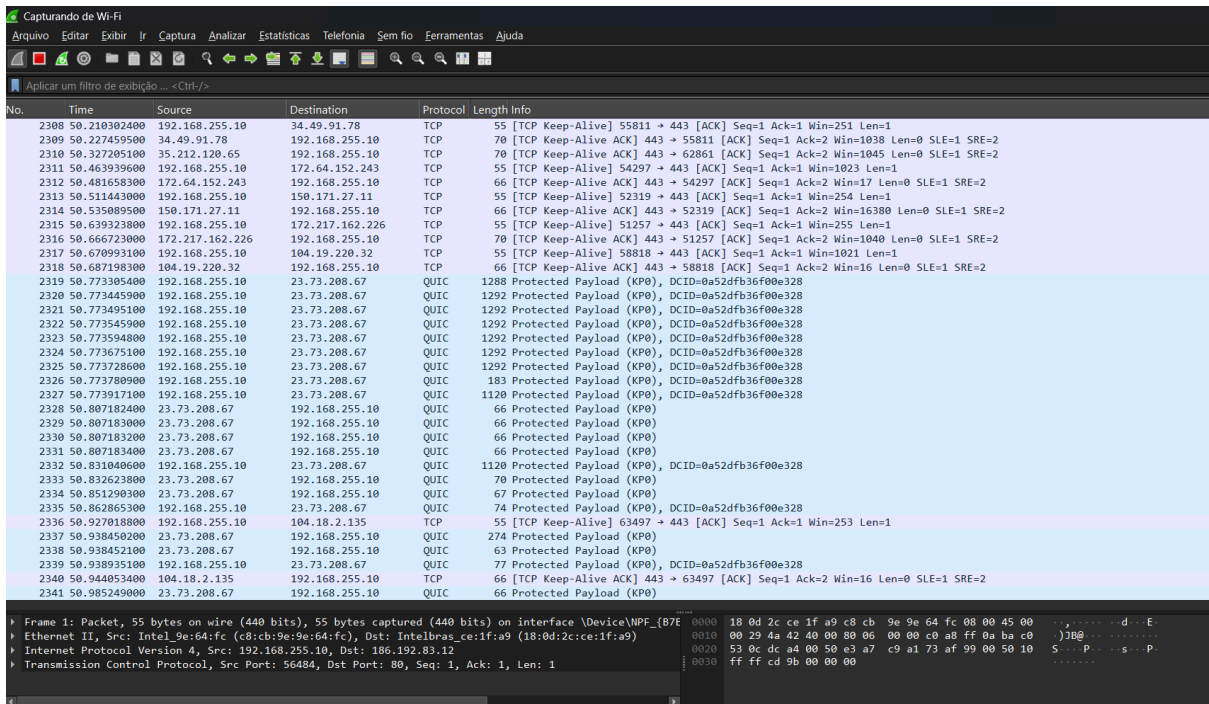
3) Abra um Navegador:

R:



4) Seleccione uma Interface:

R:



5) Tráfego Destinado ao seu IP:

Na opção de filtro, selecione ip.dst == seu-ip.

R:

Capturando de Wi-Fi

Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonía Sem fio Ferramentas Ajuda

ip.dst == 192.168.255.10

No.	Time	Source	Destination	Protocol	Length	Info
128139	2.340580400	20.209.37.1	192.168.255.10	TCP	54	443 → 56476 [FIN, ACK] Seq=10645 Ack=214744 Win=12583936 Len=0
128140	2.540581200	52.225.51.39	192.168.255.10	TCP	54	443 → 56475 [FIN, ACK] Seq=12668 Ack=2561 Win=4194816 Len=0
128144	8.118362500	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128161	9.299665600	172.217.30.78	192.168.255.10	UDP	83	443 → 59007 Len=37
128162	9.299666100	172.217.30.78	192.168.255.10	UDP	83	443 → 59007 Len=37
128165	9.514213000	142.251.155.4	192.168.255.10	TLSv1.3	131	Application Data
128167	9.537000000	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128168	9.537003900	142.251.155.4	192.168.255.10	TCP	58	443 → 61088 [ACK] Seq=7848 Ack=6893 Win=263424 Len=0
128169	9.542577000	142.251.155.4	192.168.255.10	TCP	58	443 → 61088 [FIN, ACK] Seq=7848 Ack=6893 Win=263424 Len=0
128172	9.770958000	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128173	9.854622900	52.104.168.53	192.168.255.10	TCP	54	443 → 56474 [RST, ACK] Seq=13944 Ack=3404 Win=0 Len=0
128175	10.009607200	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128177	10.246685700	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128179	10.321776800	108.177.123.84	192.168.255.10	TCP	70	[TCP Keep-Alive ACK] 443 → 50672 [ACK] Seq=6349 Ack=1908 Win=268032 Len=0 SLE=1907 SRE=1908
128181	10.421916900	52.97.48.210	192.168.255.10	TCP	66	[TCP Keep-Alive ACK] 443 → 56232 [ACK] Seq=1 Ack=2 Win=255 Len=0 SLE=1 SRE=2
128183	10.506355600	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128185	10.737789400	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128188	11.219356300	172.217.30.78	192.168.255.10	UDP	72	443 → 59007 Len=26
128189	11.227305500	172.217.30.78	192.168.255.10	UDP	73	443 → 59007 Len=27
128193	11.464470600	20.190.173.144	192.168.255.10	TCP	66	443 → 50673 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1420 WS=256 SACK_PERM
128196	11.499665900	20.190.173.144	192.168.255.10	TLSv1.3	1474	Server Hello, Change Cipher Spec
128197	11.499669300	20.190.173.144	192.168.255.10	TCP	1474	443 → 50673 [ACK] Seq=1421 Ack=480 Win=4194048 Len=1420 [TCP PDU reassembled in 128203]
128198	11.499670600	20.190.173.144	192.168.255.10	TCP	1474	443 → 50673 [ACK] Seq=2841 Ack=480 Win=4194048 Len=1420 [TCP PDU reassembled in 128203]
128199	11.499671500	20.190.173.144	192.168.255.10	TCP	1474	443 → 50673 [ACK] Seq=4261 Ack=480 Win=4194048 Len=1420 [TCP PDU reassembled in 128203]
128200	11.499673000	20.190.173.144	192.168.255.10	TCP	1474	443 → 50673 [ACK] Seq=5681 Ack=480 Win=4194048 Len=1420 [TCP PDU reassembled in 128203]
128202	11.503443600	20.190.173.144	192.168.255.10	TCP	1474	443 → 50673 [ACK] Seq=7101 Ack=480 Win=4194048 Len=1420 [TCP PDU reassembled in 128203]
128203	11.503445300	20.190.173.144	192.168.255.10	TLSv1.3	364	Application Data
128210	11.656434600	20.190.173.144	192.168.255.10	TCP	1474	[TCP SACK_RETRANSmission] 443 → 50673 [PSH, ACK] Seq=7411 Ack=480 Win=4194048 Len=1420 [TCP PDU reassembled in 128203]
128212	11.571108400	20.190.173.144	192.168.255.10	TCP	54	443 → 50673 [ACK] Seq=8831 Ack=1269 Win=4193280 Len=0
128213	11.571109800	20.190.173.144	192.168.255.10	TLSv1.3	157	Application Data
128214	11.571110200	20.190.173.144	192.168.255.10	TCP	54	443 → 50673 [ACK] Seq=8934 Ack=2689 Win=4194560 Len=0
128216	11.572092500	20.190.173.144	192.168.255.10	TCP	54	443 → 50673 [ACK] Seq=8934 Ack=4109 Win=4194560 Len=0
128217	11.612599200	20.190.173.144	192.168.255.10	TCP	54	443 → 50673 [ACK] Seq=8934 Ack=4748 Win=4193792 Len=0
128218	11.855294200	20.190.173.144	192.168.255.10	TCP	1474	443 → 50673 [ACK] Seq=8934 Ack=4748 Win=4193792 Len=1420

6) Ping em 8.8.8.8:

- Abra o terminal e execute o comando ping 8.8.8.8
- Na opção de filtro, escolha: ip.addr == seu-ip and icmp.type == 8 and ip.dst == 8.8.8.8

R:

Capturando de Wi-Fi

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ip.addr == 192.168.255.10 and icmp.type == 8 and ip.dst == 8.8.8.8

No.	Time	Source	Destination	Protocol	Length	Info
* 100255	683.618018300	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 100259)
100402	684.629228600	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 100403)
100469	685.669616800	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 100470)
100568	686.687087400	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 100569)
137361	1183.7427132..	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 137364)
137365	1184.7499867..	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 137371)
137387	1185.7634741..	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 137388)
137399	1186.7963178..	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 137401)

7) ARP:

- No filtro, selecione opção ARP

R:

Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonia Sem fio Ferramentas Ajuda						
arp						
No.	Time	Source	Destination	Protocol	Length Info	
87414	589.039712500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
87415	589.039745000	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
88342	600.049080300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
88343	600.049112700	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
89357	611.158771900	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
89358	611.158794300	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
90868	622.218332500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
90870	622.218370800	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
91887	633.078505200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
91888	633.078528000	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
93081	644.131666600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
93082	644.131683600	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
95883	655.098754800	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
95884	655.098784200	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
97131	666.147713600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
97133	666.147880100	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
98896	677.119476500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
98897	677.119504200	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
101144	688.268397900	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
101145	688.268421700	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
102625	699.154421500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
102626	699.154459600	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
103762	710.165266000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
103763	710.165288800	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
105209	721.158342200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
105210	721.158366700	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
106393	732.169820000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
106394	732.169857900	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
108408	743.222956200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
108409	743.222980700	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
108536	754.222607300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
108537	754.222639000	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
108715	765.218451200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1

8) ARP:

- O que acontece se usarmos o filtro “arp.dst.proto_ipv4 == seu-ip”?

R:

Capturando de Wi-Fi						
Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonia Sem fio Ferramentas Ajuda						
arp.dst.proto_ipv4 == 192.168.255.10						
No.	Time	Source	Destination	Protocol	Length Info	
75103	478.939184300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
76365	490.185515000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
77767	500.958766700	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
79151	512.037588700	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
80149	522.979215800	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
81543	534.061560000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
82768	545.000885700	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
83787	556.170383500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
85182	567.127504600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
86458	578.083660400	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
87414	589.039712500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
88342	600.049080300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
89357	611.158771900	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
90868	622.218332500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
91887	633.078505200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
93081	644.131666600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
95883	655.098754800	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
97131	666.147713600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
98896	677.119476500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
101144	688.268397900	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
102625	699.154421500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
103762	710.165266000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
105209	721.158342200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
106393	732.169820000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
108408	743.222956200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
108536	754.222607300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
108715	765.218451200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
108856	776.432101600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
108960	787.493174000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
109009	798.243066400	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
109052	809.302776300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
109111	820.247803600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
109927	831.420844400	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1
111277	842.480281700	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has 192.168.255.10? Tell 192.168.255.1

Exercícios

1) Ping em 8.8.8.8:

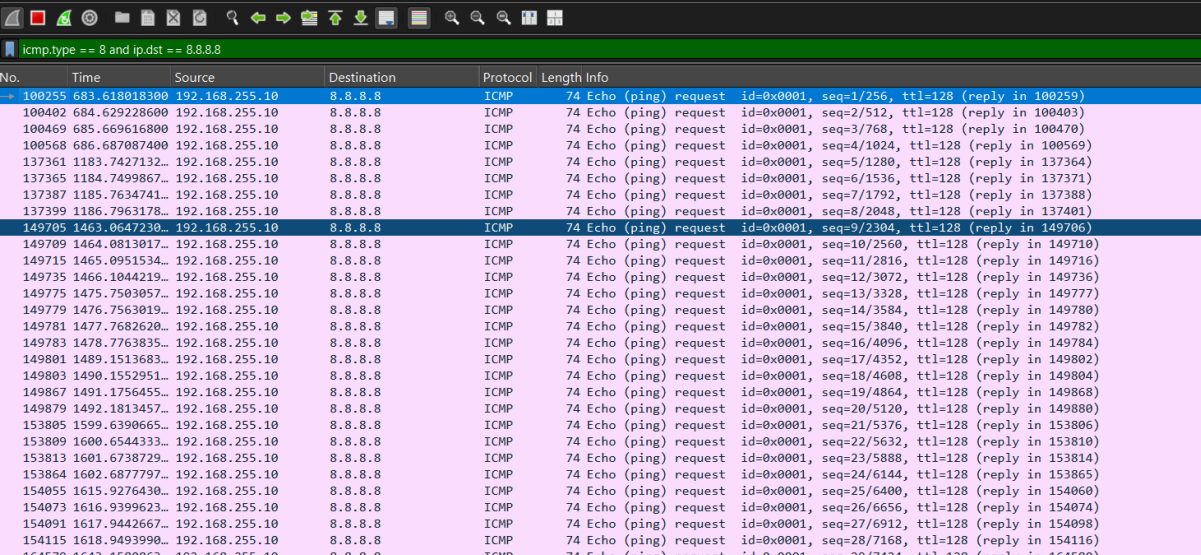
• No exercício anterior, podemos simplificar a expressão `ip.addr == 192.168.0.5 and icmp.type == 8 and ip.dst == 8.8.8.8`? Discuta sua resposta.

R: Sim, é possível simplificar a expressão. O filtro `icmp.type == 8` seleciona as mensagens ICMP do tipo Echo Request, que são as requisições enviadas pelo comando ping. Já o filtro `ip.dst == 8.8.8.8` garante que o pacote tenha como destino o endereço 8.8.8.8.

Portanto, podemos utilizar apenas:

`icmp.type == 8 and ip.dst == 8.8.8.8`

Essas duas condições são suficientes para identificar as requisições de ping destinadas ao 8.8.8.8, sem a necessidade de especificar o IP de origem.



No.	Time	Source	Destination	Protocol	Length	Info
100255	683.618018300	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 100259)
100402	684.629228600	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 100403)
100469	685.669616800	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 100470)
100568	686.687087400	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 100569)
137361	1183.7427132...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 137364)
137365	1184.7499867...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 137371)
137387	1185.7634741...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 137388)
137399	1186.7963178...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 137401)
149705	1463.0647230...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 149706)
149709	1464.0813017...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=10/2560, ttl=128 (reply in 149710)
149715	1465.0951534...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=11/2816, ttl=128 (reply in 149716)
149735	1466.1044219...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=12/3072, ttl=128 (reply in 149736)
149775	1475.7503057...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=13/3328, ttl=128 (reply in 149777)
149779	1476.7563019...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=14/3584, ttl=128 (reply in 149780)
149781	1477.7682620...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=15/3840, ttl=128 (reply in 149782)
149783	1478.7763835...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=16/4096, ttl=128 (reply in 149784)
149801	1489.1513683...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 149802)
149803	1490.1552951...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=18/4608, ttl=128 (reply in 149804)
149867	1491.1756455...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=19/4864, ttl=128 (reply in 149868)
149879	1492.1813457...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=20/5120, ttl=128 (reply in 149880)
153805	1599.6390665...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=21/5376, ttl=128 (reply in 153806)
153809	1600.6544333...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (reply in 153810)
153813	1601.6738729...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (reply in 153814)
153864	1602.6877797...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (reply in 153865)
154055	1615.9276430...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=25/6400, ttl=128 (reply in 154060)
154073	1616.9399623...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=26/6656, ttl=128 (reply in 154074)
154091	1617.9442667...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 154098)
154115	1618.9493990...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 154116)
164579	1643.1580863...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 164580)

2) ICMP:

• O que acontece se usarmos o filtro “icmp”?

R: O filtro `icmp` exibe todos os pacotes que utilizam o protocolo ICMP, incluindo tanto as mensagens Echo Request (solicitação do ping) quanto as Echo Reply (resposta do destino). Portanto, ele é mais abrangente do que o filtro `icmp.type == 8`.

Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonía Sem fio Ferramentas Ajuda									
icmp.type == 8									
No.	Time	Source	Destination	Protocol	Length	Info			
→ 100255	-1226.320715...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=1/256, ttl=128 (reply in 100259)		
100402	-1225.309505...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=2/512, ttl=128 (reply in 100403)		
100469	-1224.269117...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=3/768, ttl=128 (reply in 100470)		
100568	-1223.251646...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=4/1024, ttl=128 (reply in 100569)		
137361	-726.1960208...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=5/1280, ttl=128 (reply in 137364)		
137365	-725.1887473...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=6/1536, ttl=128 (reply in 137371)		
137387	-724.1752599...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=7/1792, ttl=128 (reply in 137388)		
137399	-723.1424162...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=8/2048, ttl=128 (reply in 137401)		
149705	-446.8740110...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=9/2304, ttl=128 (reply in 149706)		
149709	-445.8574323...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=10/2560, ttl=128 (reply in 149710)		
149715	-444.8435806...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=11/2816, ttl=128 (reply in 149716)		
149735	-443.8343121...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=12/3072, ttl=128 (reply in 149736)		
149775	-434.1884283...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=13/3328, ttl=128 (reply in 149777)		
149779	-433.1824321...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=14/3584, ttl=128 (reply in 149780)		
149781	-432.1704720...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=15/3840, ttl=128 (reply in 149782)		
149783	-431.1623505...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=16/4096, ttl=128 (reply in 149784)		
149801	-420.7873657...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=17/4352, ttl=128 (reply in 149802)		
149803	-419.7834389...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=18/4608, ttl=128 (reply in 149804)		
149867	-418.7630885...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=19/4864, ttl=128 (reply in 149868)		
149879	-417.7573883...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=20/5120, ttl=128 (reply in 149880)		
153805	-310.2996675...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=21/5376, ttl=128 (reply in 153806)		
153809	-309.2843007...	192.168.255.10	8.8.8.8	ICMP	74	Echo (ping) request	id=0x0001, seq=22/5632, ttl=128 (reply in 153810)		

3) ARP:

- Qual é a diferença de usarmos os filtros “arp.opcode == 1” e “arp.opcode == 2”?

R: A diferença está no tipo de mensagem ARP:

arp.opcode == 1: mostra ARP Request, ou seja, uma solicitação para descobrir o endereço MAC associado a um determinado endereço IP.

arp.opcode == 2: mostra ARP Reply, ou seja, a resposta informando o endereço MAC correspondente ao IP solicitado.

Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonía Sem fio Ferramentas Ajuda									
arp.opcode == 1									
No.	Time	Source	Destination	Protocol	Length	Info			
76365	490.185515000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
77767	500.958766700	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
79151	512.037588700	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
80149	522.979215800	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
81543	534.061560000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
82768	545.000885700	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
83787	556.170383500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
85182	567.127504600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
86458	578.083660400	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
87414	589.039712500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
88342	600.049080300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
89357	611.158771900	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
90868	622.218332500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
91887	633.078505200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
93081	644.131666600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
95883	655.098754800	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
97131	666.147713600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
98896	677.119476500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
101144	688.268397900	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
102625	699.154421500	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
103762	710.165266000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
105209	721.158342200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
106393	732.169820000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
108408	743.222956200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
108536	754.222607300	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
108715	765.218451200	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
108856	776.432101600	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
108960	787.493174000	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1
109009	798.243066400	Intelbras_ce:1f:a9	Intel_9e:64:fc	ARP	60	Who has	192.168.255.10?	Tell	192.168.255.1

Capturando de Wi-Fi

Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonia Sem fio Ferramentas Ajuda

arp.opcode == 2

No.	Time	Source	Destination	Protocol	Length	Info
190245	1888.2149000...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
191481	1899.2251251...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
192515	1910.2348424...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
194365	1921.2447482...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
196224	1932.2549116...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
197321	1943.2647025...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
198525	1954.2749384...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
199827	1965.2847981...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
200883	1976.3521316...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
202155	1987.3077836...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
204592	1998.3155179...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
205787	2009.3245211...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
206309	2020.3346297...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
206657	2031.3488860...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
207540	2042.3997136...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
208455	2053.4590673...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
209473	2064.5180853...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
210501	2075.4396746...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
211540	2086.6368701...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
212919	2097.4049588...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
213776	2108.4478620...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
214733	2119.5067744...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
215850	2130.5660336...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
217120	2141.4443452...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
218351	2152.4806924...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
219461	2163.5384499...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
220373	2174.4743539...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
221415	2185.5547223...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc
222464	2196.6147144...	Intel_9e:64:fc	Intelbras_ce:1f:a9	ARP	42	192.168.255.10 is at c8:cb:9e:9e:64:fc

4) DHCP:

- O que acontece se usarmos o filtro “dhcp”?

R: O filtro dhcp exibe todos os pacotes relacionados ao protocolo DHCP, permitindo visualizar a comunicação entre o computador e o servidor DHCP, como as mensagens usadas para solicitar e receber um endereço IP e outras configurações de rede.

Capturando de Wi-Fi

Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonia Sem fio Ferramentas Ajuda

dhcp

No.	Time	Source	Destination	Protocol	Length	Info
287207	2734.5023099...	192.168.255.10	192.168.255.1	DHCP	342	DHCP Release - Transaction ID 0x262151e7
287217	2736.6225280...	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x2ee6e0a5
287218	2736.6761722...	192.168.255.1	192.168.255.10	DHCP	590	DHCP Offer - Transaction ID 0x2ee6e0a5
287219	2736.6847522...	0.0.0.0	255.255.255.255	DHCP	363	DHCP Request - Transaction ID 0x2ee6e0a5
287220	2738.1684184...	0.0.0.0	255.255.255.255	DHCP	363	DHCP Request - Transaction ID 0x2ee6e0a5

5) ICMPv6:

- O que acontece se usarmos o filtro “icmpv6”?

R: O filtro icmpv6 exibe todos os pacotes que utilizam o protocolo ICMPv6, usado para comunicação e controle em redes IPv6. Ele pode mostrar mensagens como Echo Request e Echo Reply, além de mensagens de descoberta de vizinhos e roteadores.

Capturando de Wi-Fi						
Arquivo Editar Exibir Ir Captura Analizar Estatísticas Telefonía Sem fio Ferramentas Ajuda						
icmipv6						
No.	Time	Source	Destination	Protocol	Length Info	
108508	751.446469100	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
108517	751.459572500	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
111275	841.151179500	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
111276	841.457624500	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
116438	898.698953700	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
116440	899.497134900	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
127204	1018.7162337...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
127209	1019.4601829...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
127519	1034.9919332...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
127530	1035.5564044...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
136844	1164.3802333...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
136847	1164.5496758...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
138541	1216.7760212...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
138552	1216.8078096...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
143881	1275.6311453...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
143892	1275.7897601...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
147262	1322.0187423...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
147271	1322.5298673...	fe80::d8df:45ff:fe6...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
148780	1359.8048967...	fe80::6c27:9bff:fed...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
148781	1359.8048971...	fe80::6c27:9bff:fed...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287208	2734.5482437...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287209	2734.7369730...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287211	2734.9721827...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287213	2735.5750066...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287214	2735.5757004...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287215	2735.9824332...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287226	2740.7463993...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287236	2740.9798907...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287238	2741.1049858...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287239	2741.1052466...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287242	2741.1386605...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287248	2741.1500837...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287250	2741.1673933...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
287270	2741.4806273...	fe80::8438:c84c:d99...	ff02::16	ICMPv6	110	Multicast Listener Report Message v2